

Dynamic Programming & Reinforcement Learning

Value Iteration and Policy Iteration

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Content for Today

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Optional: why policy iteration converges Policy improvement, optimality, and finite termination.	45–59
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What we already know, and what we will do today

Lecture 3 already justified Bellman iteration:

$$V_{k+1} = T^{\pi} V_k \longrightarrow V^{\pi} \quad \text{fixed-policy evaluation,}$$
$$V_{k+1} = T^{*} V_k \longrightarrow V^{*} \quad \text{optimal control.}$$

It also gave Bellman-residual bounds for stopping after finitely many updates.

Today we use these results to:

- compare iterative evaluation with solving the same Bellman linear system;
- work through numerical updates, computational costs, and greedy-policy certificates;
- reuse evaluation inside **policy iteration**, then compare PI with VI.
  The improvement and convergence proofs are **optional**.

Setting and notation

- Finite, nonempty $\mathcal{X}$, and finite nonempty available-action sets $\mathcal{A}(x)$.
- The time-homogeneous joint law of (R_t, X_{t+1}), given the history and A_t, depends only on (X_t, A_t).
- $r(x, a) = \mathbb{E}[R_t \mid X_t = x, A_t = a]$; $\mathcal{P}(x' \mid x, a)$ is a next-state probability. Both are **known**.
- $|R_t| \leq R_{\max}$, $0 \leq \gamma < 1$; the state is observed.

V is a real table on $\mathcal{X}$; Q is a real table on feasible (x, a). All action sums and maxima use available actions.

t: environment time. k: algorithm iteration. A deterministic stationary policy chooses one $\pi(x) \in \mathcal{A}(x)$.

Recap: Bellman backups on state values

For a fixed stationary policy π, define

$$r^\pi(x) = \sum_a \pi(a \mid x) r(x,a), \qquad \mathcal{P}^\pi(x' \mid x) = \sum_a \pi(a \mid x) \mathcal{P}(x' \mid x, a).$$

For any candidate table V,

$$(T^\pi V)(x) = r^\pi(x) + \gamma \sum_{x'} \mathcal{P}^\pi(x' \mid x) V(x'),$$

$$(T^* V)(x) = \max_a \left[r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V(x') \right].$$

$TV = T(V)$ is a new table; $(TV)(x)$ is one entry. A **backup** is one application of this rule.

Recap: action values and greedy policies

For candidates on feasible state–action pairs,

$$(T^{\pi}Q)(x,a) = r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x,a) \sum_{a'} \pi(a' \mid x') Q(x',a'),$$

$$(T^{*}Q)(x,a) = r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x,a) \max_{a'} Q(x',a').$$

Choose $\pi_g(x;Q) \in \operatorname{argmax}_a Q(x,a)$. For state values, choose

$$\pi_g(x;V) \in \operatorname*{argmax}_a \left[r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x,a) V(x') \right].$$

$\max$ is a score; argmax is a set of actions. Fix a choice at each state.

Different approaches to finding a policy

- **Value-based:** approximate V^* or Q^*, then extract a greedy policy. Value iteration follows this route.
- **Direct policy search:** optimize policy parameters to increase expected return (e.g., policy gradient).
- **Hybrid:** evaluate the current policy, then use its values for a greedy policy update. Policy iteration follows this route.

Known model $\longrightarrow$ exact Bellman backups $\longrightarrow$ planning.

Later, sampled experience will replace parts of these exact computations.

Policy evaluation and control

Policy evaluation (prediction)

Input: an MDP and a fixed stationary policy π.
Output: V^π or Q^π.

Control (optimization)

Input: an MDP.
Output: V^*, Q^*, or an optimal policy.

Dynamic programming reuses the Bellman recursion instead of recomputing every possible future separately.

Lecture 3 already gave iterative evaluation. We now compare evaluation methods and reuse them inside policy iteration.

Why expanding all futures is inefficient

For comparison with Bellman iteration, expand the return from $X_1 = x$ under π. The expected second reward is

$$\mathbb{E}_x^\pi[R_2] = \sum_{a,x',a'} \pi(a \mid x)\mathcal{P}(x' \mid x, a)\pi(a' \mid x')r(x', a').$$

The return adds such terms at every horizon:

$$V^\pi(x) = \mathbb{E}_x^\pi\left[\sum_{t=1}^{\infty} \gamma^{t-1} R_t\right].$$

- A depth-h tree repeats many copies of the same continuation problem.
- With branching factor $b > 1$, the number of paths can grow as b^h.
- Bellman recursion stores one continuation value per state instead.

Same policy value, computed by a linear solve

Fix the state order $(x_1, \ldots, x_n)$. Store V^π, r^π as column vectors. $\mathcal{P}^\pi$ has current states as rows and next states as columns:

$$\mathcal{P}^\pi_{ij} = \mathcal{P}^\pi(x_j \mid x_i), \qquad (\mathcal{P}^\pi V)(x_i) = \sum_j \mathcal{P}^\pi_{ij} V(x_j).$$

The Bellman equation is

$$V^\pi = r^\pi + \gamma \mathcal{P}^\pi V^\pi \quad \iff \quad \boxed{(I - \gamma \mathcal{P}^\pi) V^\pi = r^\pi.}$$

There are n equations and n unknowns. In practice, use a linear solver rather than explicitly forming an inverse.

Why the policy-evaluation system is invertible

Set $A = I - \gamma \mathcal{P}^\pi$, with $0 \le \gamma < 1$. **Criterion:** a square matrix is invertible if and only if 0 is not an eigenvalue.

For $u \in \mathbb{C}^n$, use $\|u\|_\infty = \max_i |u_i|$. Nonnegative entries and row sums equal to one give

$$|(\mathcal{P}^\pi u)_i| \le \sum_j \mathcal{P}^\pi_{ij} |u_j| \le \|u\|_\infty \quad \Longrightarrow \quad \|\mathcal{P}^\pi u\|_\infty \le \|u\|_\infty .$$

Take any eigenpair $Au = \lambda u$, with $u \neq 0$. Since $u = Au + \gamma \mathcal{P}^\pi u$,

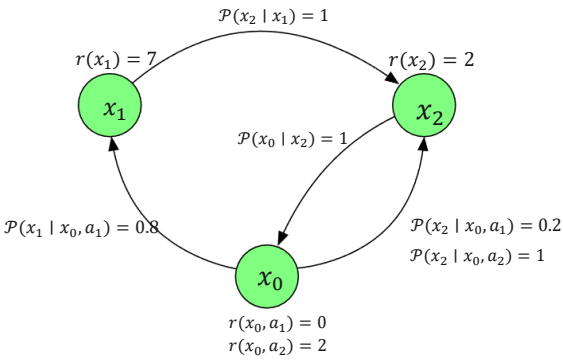
$$\begin{aligned} \|u\|_\infty &\le \|Au\|_\infty + \gamma \|\mathcal{P}^\pi u\|_\infty \\ &\le (|\lambda| + \gamma) \|u\|_\infty . \end{aligned}$$

Divide by $\|u\|_\infty > 0$: every eigenvalue satisfies

$$\boxed{|\lambda| \ge 1 - \gamma > 0.}$$

Thus A is invertible and the unique solution is V^π. Eigenvalues may be complex, so we bound their **magnitudes**.

A running MDP: all model data



State order: (x_0, x_1, x_2); $\gamma = 0.9$.
At x_0: a_1 earns 0, then moves to x_1 with probability 0.8, to x_2 with 0.2.
At x_0: a_2 earns 2, then moves to x_2 surely.

At x_1: the sole action earns 7 and moves to x_2.
At x_2: the sole action earns 2 and moves to x_0.

$\pi_1(x_0) = a_1$, $\pi_2(x_0) = a_2$; both take the sole action elsewhere. Rewards are earned on the current action; the task continues indefinitely.

Worked evaluation: the policy choosing the reward now

$\pi_2(x_0) = a_2$; sole actions at x_1, x_2; state order (x_0, x_1, x_2).

$$V^{\pi_2} = r^{\pi_2} + \gamma \mathcal{P}^{\pi_2} V^{\pi_2}, \qquad r^{\pi_2} = \begin{pmatrix} 2 \\ 7 \\ 2 \end{pmatrix}, \quad \mathcal{P}^{\pi_2} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \gamma = 0.9.$$

$$\begin{aligned} V^{\pi_2}(x_0) &= 2 + 0.9 V^{\pi_2}(x_2), \\ V^{\pi_2}(x_1) &= 7 + 0.9 V^{\pi_2}(x_2), \\ V^{\pi_2}(x_2) &= 2 + 0.9 V^{\pi_2}(x_0). \end{aligned}$$

Substitution: $V^{\pi_2}(x_0) = 3.8 + 0.81 V^{\pi_2}(x_0)$. Therefore

$$\boxed{V^{\pi_2} = (20, 25, 20)^\top.}$$

Iterating T^{π_2} must approach this same table; we will compare the computations.

From policy evaluation to control

For a fixed policy, the coefficients of T^π are known:

$$(I - \gamma \mathcal{P}^\pi) V^\pi = r^\pi.$$

For control, the maximizing action depends on the unknown value table:

$$V(x) = \max_a \left[r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V(x') \right].$$

- We cannot choose one fixed policy matrix before solving the decision problem.
- T^* is generally nonlinear, although each action score is affine in V.
- Lecture 3 already proved convergence of repeated T^* backups. We now apply that algorithm.

Value Iteration



Recap: iterative policy evaluation

Recall Lecture 3: fix π, choose any table V_0, and repeat

$$\boxed{V_{k+1} = T^\pi V_k = r^\pi + \gamma \mathcal{P}^\pi V_k.}$$

By the policy-operator contraction and its fixed point V^π,

$$\|V_{k+1} - V^\pi\|_\infty = \|T^\pi V_k - T^\pi V^\pi\|_\infty \leq \gamma \|V_k - V^\pi\|_\infty ,$$

$$\|V_k - V^\pi\|_\infty \leq \gamma^k \|V_0 - V^\pi\|_\infty \longrightarrow 0.$$

Every entry of V_{k+1} uses the **same old table** V_k. Similarly, $Q_{k+1} = T^\pi Q_k$ converges to Q^π.

Worked iterative evaluation

For π_2, use state order (x_0, x_1, x_2), $\gamma = 0.9$, and

$$r^{\pi_2} = (2, 7, 2)^\top, \quad \mathcal{P}^{\pi_2} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad V_0 = 0.$$

The Bellman rule is $V_{k+1} = r^{\pi_2} + \gamma \mathcal{P}^{\pi_2} V_k$.

$$V_{k+1} = \begin{pmatrix} 2 + 0.9 V_k(x_2) \\ 7 + 0.9 V_k(x_2) \\ 2 + 0.9 V_k(x_0) \end{pmatrix}.$$

k	$V_k(x_0)$	$V_k(x_1)$	$V_k(x_2)$
0	0	0	0
1	2	7	2
2	3.8	8.8	3.8
3	5.42	10.42	5.42

The limit is the same table as the linear solve: $V^{\pi_2} = (20, 25, 20)^\top$. It is not optimal: at x_0, switching once to a_1, then following π_2, gives $0 + 0.9[0.8(25) + 0.2(20)] = 21.6 > 20$.

Recap: value iteration for control

Recall Lecture 3: choose any real table V_0 and compute

$$\boxed{V_{k+1} = T^* V_k.}$$

Entry by entry,

$$V_{k+1}(x) = \max_a \left[r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V_k(x') \right].$$

By contraction and $T^* V^* = V^*$,

$$\|V_k - V^*\|_\infty \leq \gamma^k \|V_0 - V^*\|_\infty \longrightarrow 0.$$

After computing a candidate, extract $\pi_g(V_k)$ by one-step lookahead. A finite iterate need not equal V^*, and its greedy policy need not yet be optimal.

Synchronous value iteration

1. Initialize V_0; set $k = 0$.
2. For **every** state x and available action a, compute

$$r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V_k(x').$$

3. Store the largest score in a **new** table $V_{k+1}(x)$.
4. Compute $\delta_k = \|V_{k+1} - V_k\|_\infty$.
5. If the desired certificate holds, return V_k and $\pi_g(V_k)$. Otherwise set $k \leftarrow k + 1$ and repeat.

$\delta_k = \|T^* V_k - V_k\|_\infty$ measures the residual of V_k.

Use two arrays. Reading freshly updated entries implements a different, in-place scheme; its analysis is separate.

Worked control backup: start from zero

State order (x_0, x_1, x_2), $\gamma = 0.9$. At x_0: a_1: reward 0, next x_1/x_2 with $0.8/0.2$; a_2: reward 2, next x_2 surely. At x_1: reward 7, next x_2; at x_2: reward 2, next x_0.

$$V_{k+1}(x) = \max_a \left[r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V_k(x') \right].$$

For this model,

$$\begin{aligned} V_{k+1}(x_0) &= \max\{0.9[0.8V_k(x_1) + 0.2V_k(x_2)],\ 2 + 0.9V_k(x_2)\}, \\ V_{k+1}(x_1) &= 7 + 0.9V_k(x_2), \\ V_{k+1}(x_2) &= 2 + 0.9V_k(x_0). \end{aligned}$$

Set $V_0 = (0,0,0)^\top$. Then

$$V_1(x_0) = \max\{0, 2\} = 2, \quad V_1(x_1) = 7, \quad V_1(x_2) = 2.$$

Thus $V_1 = (2,7,2)^\top$. The action greedy with respect to V_0 at x_0 is a_2.

Quick exercise: compute the second backup

State order (x_0, x_1, x_2), $\gamma = 0.9$. At x_0: a_1: reward 0, next x_1/x_2 with $0.8/0.2$; a_2: reward 2, next x_2 surely. At x_1: reward 7, next x_2; at x_2: reward 2, next x_0.

Given $V_1 = (2, 7, 2)^\top$, compute $V_2 = T^* V_1$. Use only V_1 on the right.

$$V_{k+1}(x) = \max_a \left[r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V_k(x') \right].$$

$$\begin{aligned} V_{k+1}(x_0) &= \max\{0.9[0.8V_k(x_1) + 0.2V_k(x_2)],\ 2 + 0.9V_k(x_2)\}, \\ V_{k+1}(x_1) &= 7 + 0.9V_k(x_2), \\ V_{k+1}(x_2) &= 2 + 0.9V_k(x_0). \end{aligned}$$

1. Compute both action scores at x_0.
2. Fill $V_2 = (\ _____,\ _____,\ _____\ )^\top$.
3. Which action is $\pi_g(x_0; V_1)$? Compare with $\pi_g(x_0; V_0) = a_2$ for $V_0 = 0$, and explain the change.

Exercise solution: future rewards change the action

State order (x_0, x_1, x_2), $\gamma = 0.9$. At x_0: a_1: reward 0, next x_1/x_2 with $0.8/0.2$; a_2: reward 2, next x_2 surely. At x_1: reward 7, next x_2; at x_2: reward 2, next x_0.

Using $V_1 = (2, 7, 2)^\top$ in the same Bellman rule,

$$\begin{aligned}
\text{score}(x_0, a_1) &= 0 + 0.9[0.8(7) + 0.2(2)] = 5.4, \\
\text{score}(x_0, a_2) &= 2 + 0.9(2) = 3.8, \\
V_2(x_0) &= \max\{5.4, 3.8\} = 5.4, \\
V_2(x_1) &= 7 + 0.9(2) = 8.8, \\
V_2(x_2) &= 2 + 0.9(2) = 3.8.
\end{aligned}$$

$$\boxed{V_2 = (5.4, 8.8, 3.8)^\top, \qquad \pi_g(x_0; V_1) = a_1.}$$

The reward 7 at x_1 is now represented in the continuation table.

Repeated updates optimize longer decision horizons

Starting at $X_1 = x$, action A_t at step t earns R_t. Repeat $V_{k+1} = T^* V_k$ from $V_0 = 0$. For our model, $\gamma = 0.9$:

Steps k	Return included	$V_k(x_0)$	$V_k(x_1)$	$V_k(x_2)$
0	0	0	0	0
1	R_1	2	7	2
2	$R_1 + \gamma R_2$	5.4	8.8	3.8
3	$R_1 + \gamma R_2 + \gamma^2 R_3$	7.02	10.42	6.86

$V_k(x)$ is the optimal expected k-**step discounted return**:

$$V_k(x) = \sup_{\pi} \mathbb{E}_x^{\pi} \left[\sum_{t=1}^{k} \gamma^{t-1} R_t \right].$$

Zero and negative rewards still count as one decision step. Rewards after step k are omitted: the continuation value is zero.

Optimize over all nonanticipating policies; actions may depend on history and steps remaining.

Why one Bellman update adds one step

Induction claim: from any history ending at x, the best expected k-step discounted return is $V_k(x)$, and some rule attains it.

Base: with zero decision steps remaining, $V_0(x) = 0$.

Upper bound: choose a first action a. By the induction claim, the continuation after the next state x' is observed is at most $V_k(x')$. Thus its return is at most

$$r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V_k(x').$$

A randomized first action cannot exceed the largest such score, $(T^* V_k)(x)$.

Attainment: choose a maximizing first action, then at each observed next state use a rule attaining its k-step value. Finite actions give a maximizer; the induction claim gives the continuations.

$$\boxed{V_{k+1} = T^* V_k.}$$

The time-homogeneous Markov model makes the remaining-horizon problem depend on the next state, not on the earlier history.

Value iteration on action values

Choose Q_0 on feasible state–action pairs and repeat

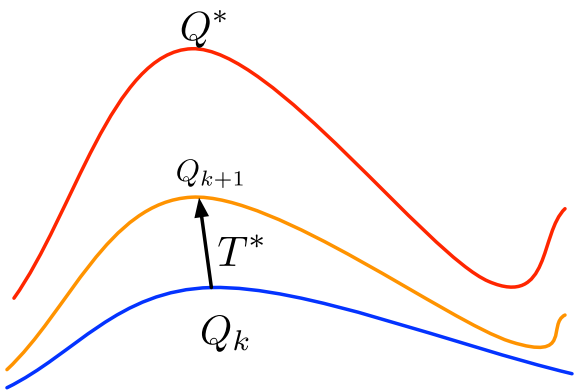
$$\boxed{Q_{k+1} = T^* Q_k,}$$

$$Q_{k+1}(x,a) = r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x,a) \max_{a'} Q_k(x',a').$$

$$\|Q_k - Q^*\|_\infty \leq \gamma^k \|Q_0 - Q^*\|_\infty \longrightarrow 0.$$

- For decisions, $\pi_g(x; Q_k) \in \operatorname{argmax}_a Q_k(x,a)$.
- The maximum is **inside** the next-state average: the next action is selected after seeing x'.
- This table stores a value for every feasible state–action pair.

One Bellman update moves a candidate

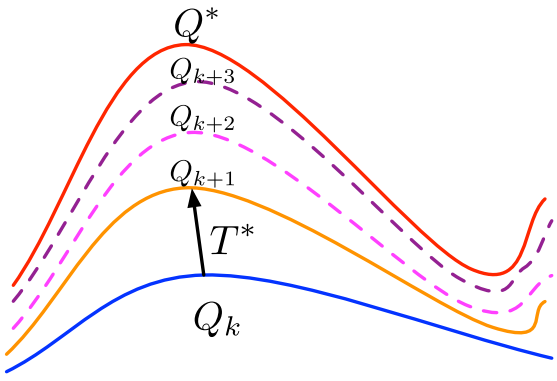


$Q_{k+1} = T^* Q_k$.
Each curve represents an entire candidate function.
The theorem guarantees

$$\|Q_{k+1} - Q^*\|_\infty \leq \gamma \|Q_k - Q^*\|_\infty .$$

The upward direction is a feature of this sketch, not a consequence of contraction alone.

Repeated updates: when values increase



If $Q_0 \leq T^* Q_0$, monotonicity gives

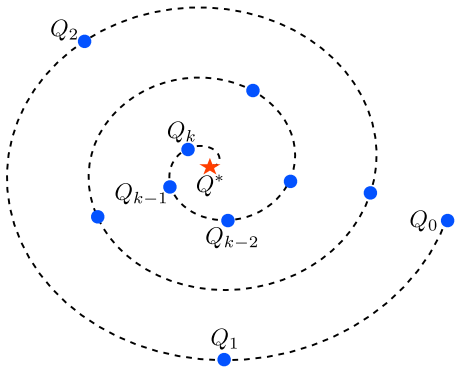
$$Q_0 \leq Q_1 \leq Q_2 \leq \cdots .$$

Indeed, $Q_k \leq Q_{k+1}$ implies

$$T^* Q_k \leq T^* Q_{k+1}.$$

Example: $Q_0 = 0$ and all mean rewards nonnegative.
Convergence still holds if this initial-order condition fails.

Contraction concerns distance to the fixed point



The points represent whole value tables.
The star is Q^*.

$$\|Q_k - Q^*\|_\infty \leq \gamma^k \|Q_0 - Q^*\|_\infty .$$

This is a schematic of shrinking distance, not an actual MDP state trajectory.
It does not require the candidates to approach from the same side or increase pointwise.

How many sweeps give a small value error?

For $V_{k+1} = T^* V_k$,

$$\|V_k - V^*\|_\infty \le \gamma^k \|V_0 - V^*\|_\infty .$$

If $V_0 = 0$ and $|R_t| \le R_{\max}$, then

$$\|V^*\|_\infty \le \sum_{t=1}^{\infty} \gamma^{t-1} R_{\max} = \frac{R_{\max}}{1-\gamma}.$$

Thus a sufficient condition for value error at most ε is

$$\gamma^k \frac{R_{\max}}{1-\gamma} \le \varepsilon.$$

For $0 < \gamma < 1$ and $0 < \varepsilon < R_{\max}/(1-\gamma)$, it suffices that

$$k \ge \left\lceil \frac{\log[R_{\max}/((1-\gamma)\varepsilon)]}{\log(1/\gamma)} \right\rceil .$$

When $\gamma = 0$, one backup is exact.

Recap: a computable certificate for value error

For any candidate V, let $\delta = \|T^*V - V\|_\infty$. Then

$$\begin{aligned}\|V - V^*\|_\infty &\le \|V - T^*V\|_\infty + \|T^*V - T^*V^*\|_\infty \\ &\le \delta + \gamma \|V - V^*\|_\infty .\end{aligned}$$

Rearranging gives

$$\boxed{\|V - V^*\|_\infty \le \frac{\delta}{1-\gamma}.}$$

For synchronous VI, $\delta_k = \|V_{k+1} - V_k\|_\infty$ certifies V_k.

Example: $\gamma = 0.9$, $\delta_k \le 0.02$ imply $\|V_k - V^*\|_\infty \le 0.2$. This is a **value-table** certificate.

A residual also certifies the greedy policy

Choose $\pi = \pi_g(V)$. Greediness gives $T^\pi V = T^* V$, so

$$\|T^\pi V - V\|_\infty = \|T^* V - V\|_\infty = \delta.$$

Apply the fixed-policy residual bound and the optimality residual bound:

$$\|V - V^\pi\|_\infty \le \frac{\delta}{1-\gamma}, \qquad \|V - V^*\|_\infty \le \frac{\delta}{1-\gamma}.$$

By the triangle inequality,

$$\boxed{\|V^* - V^\pi\|_\infty \le \frac{2\delta}{1-\gamma}.}$$

Thus the same $\gamma = 0.9$, $\delta \le 0.02$ certify policy loss at most 0.4, uniformly over starting states.

Stop according to what you want to certify

For $\varepsilon, \eta > 0$, measure $\delta_k = \|V_{k+1} - V_k\|_\infty$ after a synchronous sweep.

Goal	Definition	Stop when	Return
Value error $\leq \varepsilon$	$\|V_k - V^*\|_\infty$	$\delta_k \leq (1-\gamma)\varepsilon$	V_k
Policy loss $\leq \eta$	$\left\|V^* - V^{\pi_g(V_k)}\right\|_\infty$	$\delta_k \leq (1-\gamma)\eta/2$	$\pi_g(V_k)$

- V_k is the computed approximation to V^*.
- $V^{\pi_g(V_k)}$ is the true expected discounted return from executing the extracted policy.
- Exact optimal actions may be found before the values are exact.
- A small change at only one state does not certify the whole table: use the maximum absolute change over all states.

These bounds presume the exact model and exact backups; numerical errors must be controlled separately.

Bootstrapping: use a candidate to improve a candidate

$$\underbrace{V_k}_{\text{current continuation estimate}} \longrightarrow \underbrace{r + \gamma\,\text{expected continuation}}_{\text{one Bellman backup}} \longrightarrow V_{k+1}.$$

- This reuse of estimated continuation values is called **bootstrapping** in RL.
- It differs from resampling datasets in statistical bootstrap methods.
- VI updates a value table; it need not fully evaluate any intermediate greedy policy.
- Many learning algorithms replace exact expectations in these backups with samples.

Next: keep an explicit policy and evaluate it before changing its actions.

Policy Iteration



Alternate policy evaluation and improvement

$$\pi_k$$

$$\downarrow \quad \text{evaluate}$$

$$V^{\pi_k} \text{ or } Q^{\pi_k}$$

$$\downarrow \quad \text{choose greedy actions}$$

$$\pi_{k+1}$$

- **Evaluation:** reuse either evaluation method above, holding π_k fixed.
- **Improvement:** act greedily with respect to that policy's value.
- Policy improvement guarantees $V^{\pi_{k+1}} \geq V^{\pi_k}$ at every state; the proof is optional.

How to perform policy evaluation

For the current policy π_k, compute the **exact** fixed point

$$V^{\pi_k} = T^{\pi_k} V^{\pi_k}.$$

Options for the finite model:

- Solve $(I - \gamma \mathcal{P}^{\pi_k}) V^{\pi_k} = r^{\pi_k}$.
- From any candidate table U_0, iterate $U_{j+1} = T^{\pi_k} U_j$ to its limit. Here j indexes **inner evaluation** steps; k indexes policy changes.

After evaluating V^{π_k}, form

$$Q^{\pi_k}(x, a) = r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V^{\pi_k}(x').$$

Stopping the inner iteration early gives an approximate evaluation. The exact-PI guarantees stated here assume evaluation to the true fixed point.

Why choose the greedy action?

For a deterministic old policy π_k, choose a once at x, then follow π_k:

$$Q^{\pi_k}(x,a) = r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x,a) V^{\pi_k}(x').$$

Choose $\pi_{k+1}(x) \in \operatorname{argmax}_a Q^{\pi_k}(x,a)$. Then

$$Q^{\pi_k}(x, \pi_{k+1}(x)) = \max_a Q^{\pi_k}(x,a) \geq Q^{\pi_k}(x, \pi_k(x)) = V^{\pi_k}(x).$$

Equivalently, $T^{\pi_{k+1}} V^{\pi_k} = T^* V^{\pi_k} \geq V^{\pi_k}$.

This compares a **one-step deviation** with the old policy. The policy improvement theorem guarantees that using π_{k+1} forever is no worse; its proof is in the optional section.

Worked improvement: compare the two actions

State order (x_0, x_1, x_2), $\gamma = 0.9$. At x_0: a_1: reward 0, next x_1/x_2 with $0.8/0.2$; a_2: reward 2, next x_2 surely. At x_1: reward 7, next x_2; at x_2: reward 2, next x_0.

Old policy: $\pi_2(x_0) = a_2$, sole actions elsewhere; $V^{\pi_2} = (20, 25, 20)^\top$. Here π_1, π_2 are fixed model-policy names, not iteration indices.

$$Q^{\pi_2}(x, a) = r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V^{\pi_2}(x').$$

At x_0,

$$Q^{\pi_2}(x_0, a_1) = 0 + 0.9[0.8(25) + 0.2(20)] = 21.6,$$
$$Q^{\pi_2}(x_0, a_2) = 2 + 0.9(20) = 20.$$

The improved policy chooses a_1 at x_0: it is π_1.

21.6 is the return from choosing a_1 once, then following π_2. It is **not yet** $V^{\pi_1}(x_0)$.

Evaluate the improved policy

$\pi_1(x_0) = a_1$, sole actions elsewhere; state order (x_0, x_1, x_2).

$$V^{\pi_1} = r^{\pi_1} + \gamma \mathcal{P}^{\pi_1} V^{\pi_1}, \quad r^{\pi_1} = (0, 7, 2)^\top, \quad \mathcal{P}^{\pi_1} = \begin{pmatrix} 0 & 0.8 & 0.2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \gamma = 0.9.$$

The first row, with $V^{\pi_1}(x_1) = 7 + 0.9V^{\pi_1}(x_2)$, gives

$$V^{\pi_1}(x_0) = 0.9[0.8(7 + 0.9V^{\pi_1}(x_2)) + 0.2V^{\pi_1}(x_2)] = 5.04 + 0.828V^{\pi_1}(x_2).$$

Now use $V^{\pi_1}(x_2) = 2 + 0.9V^{\pi_1}(x_0)$:

$$V^{\pi_1}(x_0) = 5.04 + 0.828[2 + 0.9V^{\pi_1}(x_0)] = 6.696 + 0.7452V^{\pi_1}(x_0).$$

Solving $V^{\pi_1}(x_0) = 6.696/(1 - 0.7452) = 16740/637$ and substituting into the other two rows gives

$$V^{\pi_1} = \frac{1}{637}(16740, 19165, 16340)^\top \approx (26.2794, 30.0863, 25.6515)^\top.$$

Next, check whether these evaluated values suggest another policy change.

Check the improved policy's greedy actions

State order (x_0, x_1, x_2), $\gamma = 0.9$. At x_0: a_1: reward 0, next x_1/x_2 with $0.8/0.2$; a_2: reward 2, next x_2 surely. At x_1: reward 7, next x_2; at x_2: reward 2, next x_0.

The model policy π_1 chooses a_1 at x_0, and the sole actions elsewhere. Its evaluated table is

$$V^{\pi_1} = \frac{1}{637}(16740, 19165, 16340)^\top.$$

Using $Q^{\pi_1}(x, a) = r(x, a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V^{\pi_1}(x')$,

$$Q^{\pi_1}(x_0, a_1) = V^{\pi_1}(x_0) = \frac{16740}{637},$$

$$Q^{\pi_1}(x_0, a_2) = 2 + 0.9 \left(\frac{16340}{637} \right) = \frac{15980}{637}.$$

The first equality uses the action actually chosen by π_1: its Q value equals its state value.

Since $16740 > 15980$, a_1 remains greedy; other states have no alternative. No policy action changes, so

$$V^{\pi_1} = T^{\pi_1} V^{\pi_1} = T^* V^{\pi_1}.$$

By Lecture 3's unique optimality fixed point, $V^{\pi_1} = V^*$: this policy is optimal.

Exact policy iteration

1. Choose any deterministic stationary policy π_0; set $k = 0$.
2. **Evaluate:** solve $V^{\pi_k} = T^{\pi_k} V^{\pi_k}$, then compute Q^{π_k}.
3. **Improve:** for every state x, choose

$$\pi_{k+1}(x) \in \operatorname*{argmax}_a Q^{\pi_k}(x, a).$$

   Tie rule: keep $\pi_k(x)$ if it is already a maximizing action; otherwise choose a maximizer by a fixed rule.
4. If $\pi_{k+1} = \pi_k$ at all states, return π_k, V^{π_k}.
5. Otherwise increment k and repeat.

Greedy improvement also satisfies

$$T^{\pi_{k+1}} V^{\pi_k} = T^* V^{\pi_k}, \qquad T^{\pi_{k+1}} Q^{\pi_k} = T^* Q^{\pi_k}.$$

Equal values do not imply identical policies

Consider one state x and two actions a, b: both earn 0 and return to x, with $0 \leq \gamma < 1$.

$$V^{\pi_a}(x) = V^{\pi_b}(x) = 0, \qquad Q^{\pi_a}(x, a) = Q^{\pi_a}(x, b) = 0.$$

Both policies are optimal. An arbitrary tie choice could alternate

$$\pi_a, \pi_b, \pi_a, \pi_b, \ldots$$

without ever passing the policy-equality stopping test.

Keeping the old action when it is a maximizer makes this algorithm stop. Finite policy space alone does not rule out arbitrary tie cycling.

Approximate policy iteration: a preview

Exact PI uses Q^{π_k}. In practice, we may have an estimate $\widehat{Q}_k$:

$$\widehat{Q}_k \approx T^{\pi_k}\widehat{Q}_k, \qquad \pi_{k+1}(x) \in \operatorname*{argmax}_a \widehat{Q}_k(x,a).$$

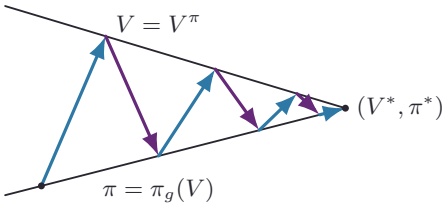
- Evaluation may stop early, use samples, or use function approximation.
- Small errors can reverse the ranking of nearly tied actions.
- The exact policy-improvement theorem cannot be applied without checking its assumptions.
- Later lectures study how approximation errors propagate.

The exact-PI guarantees assume **exact** evaluation and greedy improvement. Approximate methods need their own error analysis.

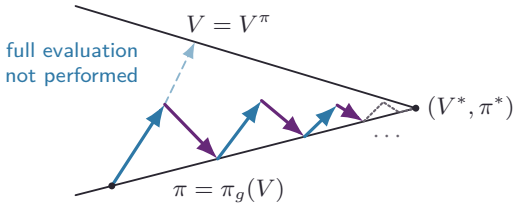
Value iteration and policy iteration: summary

Blue: update V, hold π fixed. **Purple:** choose greedy π, hold V fixed.

Policy iteration: full evaluation



Value iteration: one backup



Evaluate: $V^{\pi_k} = T^{\pi_k} V^{\pi_k}$.

Improve: $\pi_{k+1} = \pi_g(V^{\pi_k})$.

Nondecreasing policy values; finite optimal termination when old tied maximizers are kept.

Greedy policy: $\pi_{k+1} = \pi_g(V_k)$.

One backup: $V_{k+1} = T^{\pi_{k+1}} V_k = T^* V_k$.

$V_k \to V^*$; a small Bellman residual certifies value error and greedy-policy loss.

Conceptual pairs (V, π): the lines show evaluated and greedy relations. Arrow lengths do not measure error or runtime; VI's greedy policy need not converge when optimal actions tie.

Core ends here. PI improvement and termination proofs are optional.

Optional: Why Policy Iteration Converges



The policy improvement theorem

Let π be stationary, and let $\pi' = \pi_g(Q^\pi)$ be a deterministic greedy policy. Then

$$T^{\pi'}Q^\pi = T^*Q^\pi \quad \Longrightarrow \quad \boxed{Q^{\pi'} \geq T^*Q^\pi \geq Q^\pi.}$$

Every inequality is pointwise over feasible (x, a).

Proof plan:

1. One new-policy backup cannot lower Q^π.
2. Monotonicity preserves the inequality through repeated backups.
3. Contraction makes the repeated backups converge to $Q^{\pi'}$.

This uses **policy evaluation**, **greedy equality**, **monotonicity**, and **contraction**.

Improvement proof, step 1: one backup

At every feasible (x, a),

$$
\begin{aligned}
(T^{\pi'} Q^\pi)(x,a) &= (T^* Q^\pi)(x,a) && \text{greediness} \\
&= r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) \max_{a'} Q^\pi(x', a') \\
&\geq r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) \sum_{a'} \pi(a' \mid x') Q^\pi(x', a') \\
&= (T^\pi Q^\pi)(x,a) = Q^\pi(x,a). && \text{policy evaluation}
\end{aligned}
$$

A maximum is at least an average of the same action scores. Probability weights and γ are nonnegative.

$$\boxed{T^{\pi'} Q^\pi = T^* Q^\pi \geq Q^\pi.}$$

Improvement proof, step 2: repeated backups

We have $T^{\pi'}Q^{\pi} \geq Q^{\pi}$. Applying the **same** operator $T^{\pi'}$ and using monotonicity gives

$$(T^{\pi'})^2 Q^{\pi} \geq T^{\pi'}Q^{\pi} = T^*Q^{\pi} \geq Q^{\pi}.$$

Repeat inductively. For every integer $m \geq 1$,

$$\boxed{(T^{\pi'})^m Q^{\pi} \geq T^*Q^{\pi} \geq Q^{\pi}.}$$

- $(T^{\pi'})^m$ means m operator applications.
- π' remains fixed while m grows.
- These are **evaluation** iterates of the new policy, initialized at the old policy's values.

Improvement proof, step 3: take the limit

The policy operator is a γ-contraction with fixed point $Q^{\pi'}$:

$$\left\| (T^{\pi'})^m Q^\pi - Q^{\pi'} \right\|_\infty \leq \gamma^m \left\| Q^\pi - Q^{\pi'} \right\|_\infty \longrightarrow 0.$$

Each iterate satisfies

$$(T^{\pi'})^m Q^\pi \geq T^* Q^\pi \geq Q^\pi.$$

Uniform convergence implies convergence at every feasible pair. Take the limit of each component:

$$\boxed{Q^{\pi'} \geq T^* Q^\pi \geq Q^\pi.}$$

The theorem now concerns following π' forever.

The state-value form of improvement

For $\pi' = \pi_g(V^\pi)$,

$$(T^{\pi'} V^\pi)(x) = \max_a Q^\pi(x,a) \geq \sum_a \pi(a \mid x) Q^\pi(x,a) = V^\pi(x).$$

For $m \geq 1$, monotonicity propagates this first inequality:

$$(T^{\pi'})^m V^\pi \geq T^{\pi'} V^\pi = T^* V^\pi \geq V^\pi.$$

The left side converges to $V^{\pi'}$ by policy-operator contraction. Hence

$$\boxed{V^{\pi'} \geq T^* V^\pi \geq V^\pi.}$$

If the old deterministic action is **not** a maximizer at a state x, then

$$(T^* V^\pi)(x) > V^\pi(x) \quad \Longrightarrow \quad V^{\pi'}(x) > V^\pi(x).$$

Improvement can be strict at some states and equal at others. Strictness at **every** state is not required.

What exact policy iteration guarantees

For the exact PI sequence with deterministic stationary policies:

1. $V^{\pi_{k+1}} \geq V^{\pi_k}$ pointwise.
2. $V^{\pi_{k+1}} = V^{\pi_k}$ as **whole tables** if and only if $V^{\pi_k} = V^*$.
3. $\|V^{\pi_k} - V^*\|_\infty \to 0$.
4. With finite state/action sets and the stated tie rule, the policy-equality algorithm terminates after finitely many policy changes.

strict value improvement somewhere or already optimal.

The unique optimal **value** need not have a unique optimal **policy**.

A value plateau is an optimality certificate

Suppose $V^{\pi_{k+1}} = V^{\pi_k}$ as functions. Write the common table as W.

$$\begin{aligned} W &= T^{\pi_{k+1}} W && \text{new-policy Bellman equation} \\ &= T^{\pi_{k+1}} V^{\pi_k} && W = V^{\pi_k} \\ &= T^* V^{\pi_k} && \text{greedy improvement} \\ &= T^* W. && W = V^{\pi_k} \end{aligned}$$

Thus W is a fixed point of T^*. Its unique fixed point is V^*, so

$$\boxed{V^{\pi_k} = V^{\pi_{k+1}} = V^*.}$$

Here W is only a name for the shared table, not a new assumption.

At the optimum, greedy improvement preserves value

If $V^{\pi_k} = V^*$, then

$$T^{\pi_{k+1}} V^* = T^* V^* = V^*.$$

By the unique fixed point of the new-policy operator,

$$V^{\pi_{k+1}} = V^* = V^{\pi_k}.$$

Therefore,

$$\boxed{V^{\pi_{k+1}} = V^{\pi_k} \quad \Longleftrightarrow \quad V^{\pi_k} = V^*.}$$

Why does the tie rule stop here? At every state, the incumbent action has score

$$Q^{\pi_k}(x, \pi_k(x)) = V^{\pi_k}(x) = V^*(x) = \max_a Q^{\pi_k}(x, a).$$

The last equality uses $\max_a Q^{\pi_k}(x, a) = (T^* V^{\pi_k})(x)$. Thus every incumbent action is a maximizer and is retained.

Compare policy iteration with Bellman iterates

Policy improvement gives $Q^{\pi_{k+1}} \geq T^* Q^{\pi_k}$. By monotonicity of T^*,

$$
\begin{aligned}
&Q^{\pi_1} \geq T^* Q^{\pi_0}, \\
&Q^{\pi_2} \geq T^* Q^{\pi_1} \geq (T^*)^2 Q^{\pi_0}, \\
&Q^{\pi_k} \geq (T^*)^k Q^{\pi_0} \qquad (k \geq 0).
\end{aligned}
$$

Every policy's action values are bounded above by optimal action values:

$$\boxed{Q^* \geq Q^{\pi_k} \geq (T^*)^k Q^{\pi_0}.}$$

These are three entire tables. The lower table is the k-step **VI iterate** initialized at Q^{π_0}; the middle table is a fully evaluated **PI policy**.

Convergence of action values

The sandwich holds at every pair, so

$$\begin{aligned} 0 &\le Q^*(x,a) - Q^{\pi_k}(x,a) \\ &\le Q^*(x,a) - ((T^*)^k Q^{\pi_0})(x,a). \end{aligned}$$

The gaps are nonnegative at every pair. Taking their maximum gives the first bound; optimality-operator contraction gives the second:

$$\begin{aligned} \|Q^{\pi_k} - Q^*\|_\infty &\le \left\|(T^*)^k Q^{\pi_0} - Q^*\right\|_\infty \\ &\le \gamma^k \|Q^{\pi_0} - Q^*\|_\infty \longrightarrow 0. \end{aligned}$$

PI values therefore converge uniformly to Q^*, because the lower VI trajectory does.

Next we must account for **which action the policy selects** to obtain state-value convergence.

Convergence of state values

The deterministic new policy is greedy in the **old** table Q^{π_k}. At every state,

$$\begin{aligned} V^{\pi_{k+1}}(x) &= Q^{\pi_{k+1}}(x, \pi_{k+1}(x)) \\ &\geq Q^{\pi_k}(x, \pi_{k+1}(x)) && \text{Q improvement} \\ &= \max_a Q^{\pi_k}(x, a). && \text{greedy choice} \end{aligned}$$

Use $V^*(x) = \max_a Q^*(x, a)$ and the maximum-difference bound:

$$\begin{aligned} 0 \leq V^*(x) - V^{\pi_{k+1}}(x) &\leq \max_a Q^*(x, a) - \max_a Q^{\pi_k}(x, a) \\ &\leq \max_a |Q^*(x, a) - Q^{\pi_k}(x, a)| \\ &\leq \|Q^* - Q^{\pi_k}\|_\infty . \end{aligned}$$

The right side is independent of x and tends to zero. Therefore $\|V^{\pi_{k+1}} - V^*\|_\infty \to 0$.

Why the algorithm terminates in finite time

The number of deterministic stationary policies is

$$N = \prod_{x \in \mathcal{X}} |\mathcal{A}(x)| \quad \text{(or } |\mathcal{A}|^{|\mathcal{X}|} \text{ with a common action set)}.$$

1. If a policy is suboptimal, whole-table equality is impossible: some state strictly improves.
2. A strict value improvement cannot revisit any earlier policy, since that policy has the same fixed value table as before.
3. Only N deterministic policies are available. Hence only finitely many strict improvements can occur.
4. At an optimal policy, the old actions are maximizers. The tie rule retains them, so the algorithm stops.

At most N distinct policies are evaluated. This is a finiteness argument, not a practical efficiency guarantee.

Check your understanding

For a finite discounted MDP and **exact** computations:

1. VI has $\gamma = 0.9$ and $\|V_{k+1} - V_k\|_\infty \le 0.02$. What value error is certified, and for which table?
2. For exact PI, $V^{\pi_{k+1}}(x_0) = V^{\pi_k}(x_0)$ at one state. Does this certify optimality?
3. The policy changes only between tied maximizing actions. Must values strictly improve?
4. Why is one-step greedy improvement enough to justify following the new policy forever?

Useful facts:

$$\|V - V^*\|_\infty \le \frac{\|T^*V - V\|_\infty}{1-\gamma}, \qquad V^{\pi'} \ge T^*V^\pi \ge V^\pi.$$

Check your understanding: answers

1. $\|V_k - V^*\|_\infty \leq 0.02/(1-0.9) = 0.2$. The measured residual belongs to V_k.
2. No. The plateau theorem requires equality of the **entire** value table. Other states may still improve.
3. No. Distinct tied-action policies can have identical values. Our tie rule avoids such changes.
4. Greediness gives $T^{\pi'} V^\pi \geq V^\pi$. Monotonicity preserves the inequality through repeated $T^{\pi'}$ updates; contraction makes their limit $V^{\pi'}$.

one-step inequality $\xrightarrow{\text{monotonicity}}$ all evaluation iterates $\xrightarrow{\text{contraction}}$ true new-policy value.

Optional: Further Planning Methods



Modified policy iteration

Maintain a candidate V_k, choose $\pi_{k+1} = \pi_g(V_k)$, and apply $m \geq 1$ evaluation backups:

$$\boxed{V_{k+1} = (T^{\pi_{k+1}})^m V_k.}$$

- If $m = 1$, greediness gives $V_{k+1} = T^* V_k$: ordinary VI.
- If evaluation continues to its limit, $V_{k+1} = V^{\pi_{k+1}}$: a full PI evaluation step.
- Finite $m > 1$ trades more work per policy change for a more accurate evaluation.

This is **modified policy iteration**. Its convergence needs an analysis of these partial-evaluation updates; the exact-PI proof above does not apply verbatim.

Approximation error can reverse a small action gap

For π_2 in the running model,

$$Q^{\pi_2}(x_0, a_1) = 21.6, \qquad Q^{\pi_2}(x_0, a_2) = 20.$$

If $\left\| \widehat{Q} - Q^{\pi_2} \right\|_\infty \leq e$, then

$$\widehat{Q}(x_0, a_1) - \widehat{Q}(x_0, a_2) \geq (21.6 - 20) - 2e = 1.6 - 2e.$$

- If $e < 0.8$, the action ranking is guaranteed correct.
- If $e = 1$, estimates 20.6 and 21 are allowed and reverse the ranking.

At a state with at least two actions, let g be the best score minus the second-best score. If $g > 2e$, the unique best action remains best under uniform Q error e.

Greedy-policy loss from value-table error

Let $\pi = \pi_g(V)$, $e = \|V - V^*\|_\infty$, and $D = \|V^* - V^\pi\|_\infty$. The symbols e, D name value error and policy loss. Since $T^\pi V = T^* V$,

$$\begin{aligned} D &= \|T^* V^* - T^\pi V^\pi\|_\infty \\ &\le \|T^* V^* - T^* V\|_\infty + \|T^\pi V - T^\pi V^*\|_\infty + \|T^\pi V^* - T^\pi V^\pi\|_\infty \\ &\le 2\gamma e + \gamma D. \end{aligned}$$

Thus

$$\boxed{\left\|V^* - V^{\pi_g(V)}\right\|_\infty \le \frac{2\gamma}{1-\gamma} \|V - V^*\|_\infty .}$$

Combining with $e \le \delta/(1-\gamma)$ gives $2\gamma\delta/(1-\gamma)^2$. The direct residual certificate in the core may be tighter.

Beyond finite policy counting

For n states and at most m actions per state, policy counting gives at most m^n deterministic rules. For example, 100 states with 4 actions each give $4^{100} \approx 1.6 \times 10^{60}$ rules.

For exact all-state greedy (Howard) PI, stronger results give a polynomial bound of the form

$$O\left(\frac{nm}{1-\gamma}\left[1+\log\frac{1}{1-\gamma}\right]\right)$$

on policy-improvement iterations.

- For a fixed discount below one, this is polynomial in the table size.
- Dependence on $(1-\gamma)^{-1}$ matters as the discount approaches one.
- This counts **outer iterations**, not the cost of each linear solve.

This stronger theorem is stated for perspective; its proof is outside the core.

Another route: Bellman upper bounds

Define the set of tables that dominate their own optimality backup:

$$\mathcal{C} = \{V : V \geq T^* V\}.$$

- The inequality is pointwise over all states.
- $V^* \in \mathcal{C}$, because $V^* = T^* V^*$.
- We will show every $V \in \mathcal{C}$ satisfies $V \geq V^*$.

Thus $\mathcal{C}$ supplies a family of upper-bound tables.

The smallest one can be selected by an optimization problem.

Every Bellman upper-bound table dominates the optimum

If $V \in \mathcal{C}$, then $V \geq T^*V$. Monotonicity gives

$$T^*V \geq (T^*)^2 V,$$

and inductively,

$$V \geq T^*V \geq (T^*)^2 V \geq \cdots \geq (T^*)^j V \qquad (j \geq 1).$$

Contraction makes $(T^*)^j V \to V^*$. Take each component's limit:

$$\boxed{V \geq V^*.}$$

Therefore V^* is the pointwise smallest element of $\mathcal{C}$.

Choose the smallest feasible table

Choose a vector μ with $\mu(x) > 0$ at every state and solve

$$\min_{V \in \mathcal{C}} \ \mu^\top V = \min_{V \in \mathcal{C}} \sum_x \mu(x) V(x).$$

Since every feasible $V \geq V^*$,

$$\mu^\top V \geq \mu^\top V^*.$$

If $V \neq V^*$, at least one component is strictly greater; its positive weight makes the objective strictly greater. Hence V^* is the unique minimizer.

μ need not be a probability distribution. Strict positivity ensures that no state is ignored.

The Bellman inequalities are linear constraints

For each state x,

$$V(x) \geq \max_a \left[r(x,a) + \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V(x') \right]$$

holds if and only if, for **every** available action a,

$$V(x) - \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V(x') \geq r(x,a).$$

Therefore solve the linear program

$$\begin{aligned} \min_V \quad & \sum_x \mu(x) V(x) \\ \text{s.t.} \quad & V(x) - \gamma \sum_{x'} \mathcal{P}(x' \mid x, a) V(x') \geq r(x,a), \\ & \qquad\qquad \text{for every feasible } (x,a). \end{aligned}$$

There are $|\mathcal{X}|$ unrestricted real variables and $M = \sum_x |\mathcal{A}(x)|$ inequalities.

Three planning routes, one Bellman structure

- **VI:** iterate the optimality operator to its fixed point.
- **PI:** alternate exact evaluation and greedy policy improvement.
- **LP:** choose the smallest Bellman upper-bound table.

$$\text{known model} \quad + \quad \text{Bellman recursion} \quad \longrightarrow \quad \text{planning algorithms.}$$

When the model is unknown, expectations, evaluations, and comparisons must be estimated from experience.

The guarantees then require accounting for sampling and approximation error.